University of Florida, Department of Mathematics

Analysis PhD Exam, June 1996

Remark. In the sequel the word measure means positive, countably additive measure, unless otherwise stated.

Instructions. Do all of the following problems.

1. Let $f_n : \mathbb{R} \to \mathbb{R}$ be a sequence of Lebesgue-measurable functions which converges pointwise to a function g. Prove, directly from the definition of measurability, that g is measurable.

2. For $f \in L^1(\mathbb{R})$, let f_t denote the translation of f by t, i.e. $f_t(x) := f(x-t)$. Show that the map $t \mapsto f_t$ is a continuous map from $\mathbb{R}$ to $L^1(\mathbb{R})$.

3. Let μ denote the Lebesgue measure on $\mathbb{R}$. Let $T : \mathbb{R} \to \mathbb{R}$ be defined by $T(x) = x^3 - x$. Define a Borel measure ν setting $\nu(A) := \mu(T^{-1}A)$. Compute the Radon–Nikodym derivative $\frac{d\nu}{d\mu}$.

4. Let $K_n \subset [0,1]$ be a Cantor set of Lebesgue measure larger than $1 - \frac{1}{n}$. Let $V := \bigcup_{n=1}^{\infty} K_n$. Prove that, for any null set N, the set $V \setminus N$ is not a G_δ set. [Hint: Baire Category Theorem]

5. Let μ and ν be σ-finite measures on the measurable space $(X, \mathcal{B})$. Prove, from first principles, that ν can be written as $\nu = \nu_{\text{ac}} + \nu_{\text{sing}}$, with $\nu_{\text{ac}} \ll \mu$ and $\nu_{\text{sing}} \perp \mu$.

6. Let x_n be a sequence in a Banach space B. Assume that the sequence x_n converges weakly to x.

(a) Show that x is the only weak limit of the sequence x_n.

(b) Show that $\sup_n \|x_n\| < \infty$. [Hint for (b): Consider the x_n as elements of the double dual B^{**}.]

7. Let $K : [0, 2\pi] \to \mathbb{R}$ be defined by $K(x) = \frac{1}{4}\sin(x)$. Show that for every $h \in L^p([0, 2\pi])$, there exists a unique solution f to the equation

$$f + f * K = h.$$

If h is C^∞, is it true that the solution f is C^∞? Justify your answer.

8. A distribution T is called harmonic if $\Delta T = 0$, where $\Delta = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$. Show that if T is a harmonic tempered distribution, then T is a polynomial. [Hint: a distribution with support $\{0\}$ is a finite sum of multiples of Dirac's δ and its derivatives.]